Chunwei Xing

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Github Profile | Linkedin Profile | Personal Homepage

EDUCATION

University of Pennsylvania (UPenn) — Philadelphia, US
PhD. in Electrical and Systems Engineering | advised by Prof. Antonio Loquercio Aug. 2024 - Present

Eidgenössische Technische Hochschule Zürich (ETHz) — Zürich, Switzerland
MSc. in Robotics, Systems and Control | **GPA:** 5.73/6.0 (**Thesis**: 6.0/6.0) — Sep. 2020 - Nov. 2023

Tsinghua University (THU) — Beijing, China
BSc. in Mechanical Engineering | **GPA:** 3.68/4.0 (**Thesis**: 4.0/4.0), **Rank:** 12/107. — Aug. 2016 - Jul. 2020

Delft University of Technology (TUD) — Delft, Netherlands
Exchange Program in Mechanical Engineering | **GPA:** 8.81/10.0 — Sep. 2018 - Feb. 2019

PUBLICATIONS

- J. Xing, L. Bauersfeld, Y. Song, **Xing, Chunwei**, and D. Scaramuzza. Contrastive learning for enhancing robust scene transfer in vision-based agile flight. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 5330–5337, 2024
- **C. Xing**, X. Sun, A. Cramariuc, S. Gull, J. J. Chung, C. Cadena, R. Siegwart, and F. Tschopp. Descriptellation: Deep Learned Constellation Descriptors for SLAM. *arXiv preprint arXiv:2203.00567*, 2022

RESEARCH EXPERIENCE

Vision-Based Autonomous Drone Racing using Imitation Learning — Zürich, Switzerland
Robotics and Perception Group (RPG), UZH | Thesis Student/Research Assistant | `PyTorch` — Dec. 2022 - Present
- Developed the **1st** learning-based policy using **only** perceptual inputs **without** IMU and deployed it in the real world.
- Explored and benchmarked controller performances against various learning-based visual representations.
- Improved learning efficiency (**4x** faster) and performed real-world agile flight (max speed ~8.0 m/s).

Learning-Based Pose Selection for Path Planning on Challenging Terrains — Zürich, Switzerland
Robotic Systems Lab (RSL), ETHz | Project Student | `PyTorch/CMake/C++` — Oct. 2021 - Feb. 2022
- Developed a base pose selector and trained *RL* policies for a wheeled-legged robot to ground on challenging terrains.
- Achieved a success rate of *80%* at a grounding error of *0.05* m on different challenging terrains, e.g. Holes, and Steps.
- Integrated the pose selector into the global planner (RRT*) and benchmarked it against brute force search algorithm.

Descriptellation: Deep Learned Constellation Descriptors — Zürich, Switzerland
Autonomous Systems Lab (ASL), ETHz | Project Student | `PyTorch/Open3D` — Feb. 2021 - Jun. 2021
- Implemented a graph-based learned descriptor of fused geometric and semantic information extracted from object constellations by introducing semantic embedding and *graph attention* layers into DeepGCNs.
- Our descriptor outperformed SOTA learning-based and handcrafted descriptors in real-world datasets, such as GOSMatch Vertex in global localization.

INTERNSHIP

LLM Research Engineer — Beijing, China
BOSS Zhipin Lab | PyTorch | — Feb. 2024 - Aug. 2024
- Developed a probabilistic algorithm to minimize the risks of LLM agents generating wrong answers in online chats.
- Working on LLMs' fine-tuning with DPO, PEFT(LoRA, soft-prompt tuning), and inference optimization with vLLM.

Machine Learning Engineer Intern — Baden, Switzerland
ABB Corporate Research Center | PyTorch/CMake/C++/InfluxDB | — May. 2022 - Nov. 2022
- Developed a generalized testing framework for benchmarking real-time performance on Linux with preempt-rt.
- Proposed and implemented *Physics-Informed Neural Networks* to predict execution time distributions.

LANGUAGE & SKILLS

- Chinese(Mandarin): Native. German: Beginner(A2). English: Duolinguo(135), TOEFL(108, S24), GRE(322)
- Machine Learning/Robotics: TensorFlow, Keras, PyTorch, TensorRT, CUDA, Scikit-learn, OpenCV, Open3D, ROS
- Programming Languages: C, C#, C++, Python, Linux Bash, MATLAB, LaTex, React
- Softwares/Tools: Docker, Git, CMake, Unity, Blender

ACHIEVEMENTS

- National Scholarship for Encouragement — *Nov.2019, Nov.2018, Nov.2017*